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Worksheet · Energy efficiency

Power Factor Self-Check

Work out your power factor from your electricity bill, estimate what it costs you each year, and know what to ask before you buy correction equipment.

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1 Find your power factor on the bill

Look for kWh (active energy) and kVArh (reactive energy) on your monthly account. Eskom time-of-use accounts show both; many municipal large-user accounts show kVA demand instead. Copy what your account shows.

Power factor = kWh ÷ √(kWh² + kVArh²)

Month	kWh	kVArh	Power factor	Peak demand (kVA)	Reactive charge (R)

Quick check

If kVArh is more than 30% of kWh, your power factor is 0.96 or lower. On Eskom's Megaflex, Miniflex and Ruraflex tariffs, that excess is charged in June, July and August. Include those three months above if you have them.

2 Estimate what your kVA demand costs

If your tariff bills demand or network capacity in kVA, a better power factor lowers that charge every month. Fill in the blanks.

A. Peak real demand (kW). Use kVA × power factor if kW isn't shown	
B. Current power factor at peak	
C. kVA today = A ÷ B	
D. kVA at 0.98 = A ÷ 0.98	
E. kVA saved = C − D	
F. Your demand + network charge (R per kVA per month)	
G. Saving per year = E × F × 12	

3 Estimate the Eskom winter reactive charge

Skip this step if you're not on an Eskom time-of-use tariff. Use a winter month (June to August) and the kWh and kVArh recorded in peak and standard periods.

H. kWh in peak and standard periods	
I. kVArh in peak and standard periods	
J. Free allowance = H × 0.30	
K. Chargeable kVArh = I − J (zero if negative)	
L. Your c/kVArh rate (Megaflex 2026/27: 34.49c excl. VAT)	
M. Charge per winter month = K × L ÷ 100	

Eskom works out the excess per 30-minute period on Megaflex, so this monthly estimate is a guide. Your bill shows the actual figure.

4 Walk the site

If you already have a correction bank

- [ ] The controller shows a power factor of 0.97 or higher during working hours
- [ ] No alarms on the controller, and all steps switch in when the load is high
- [ ] The cabinet isn't hot to the touch and its fans run
- [ ] Capacitors aren't bulging or leaking, and contactors aren't pitted or noisy
- [ ] It has been serviced in the last 12 months

Loads that make correction harder

- [ ] Variable speed drives on pumps, fans or compressors
- [ ] UPS systems, large banks of LED lighting, or data equipment
- [ ] Welding sets, induction furnaces or other loads that change quickly
- [ ] A solar PV system, installed or planned

If you ticked any of these, ask for a harmonic measurement before buying a capacitor bank. A detuned bank or active filter may be needed.

5 Questions to ask a correction supplier

- [ ] Was the bank sized on at least a week of logged data at our incomer, or on our bills?
- [ ] What harmonic distortion did you measure, and is the bank detuned to suit it?
- [ ] What power factor will we reach at peak demand, and at light load at night?
- [ ] How many steps does the bank have, and how does it avoid overcorrection?
- [ ] Have you allowed for our solar system, existing or planned?
- [ ] What saving per month do you expect, and how will we verify it on the bill?
- [ ] What maintenance does it need, how often, and who will do it?

Rules of thumb

- Aim for 0.97 to 0.99, not 1.00. A leading power factor at light load can raise voltages.
- Correction mainly saves on kVA and reactive charges. It saves only a small amount of kWh through lower cable and transformer losses.
- Check power factor again after any solar installation or major new load.

Want it measured properly?

Every Bynergex Energy Loss Audit logs your power factor, kVAr demand and harmonics at the incomer, reads your tariff, and tells you what correction is worth and what type you need. We don't sell correction equipment, so the advice is independent.

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General guidance only, not a substitute for a site measurement. Tariff details were checked against Eskom's 2026/27 Tariffs and Charges Booklet in October 2026; tariffs change each year, so use the figures on your own account. © 2026 Bynergex Consulting (Pty) Ltd.